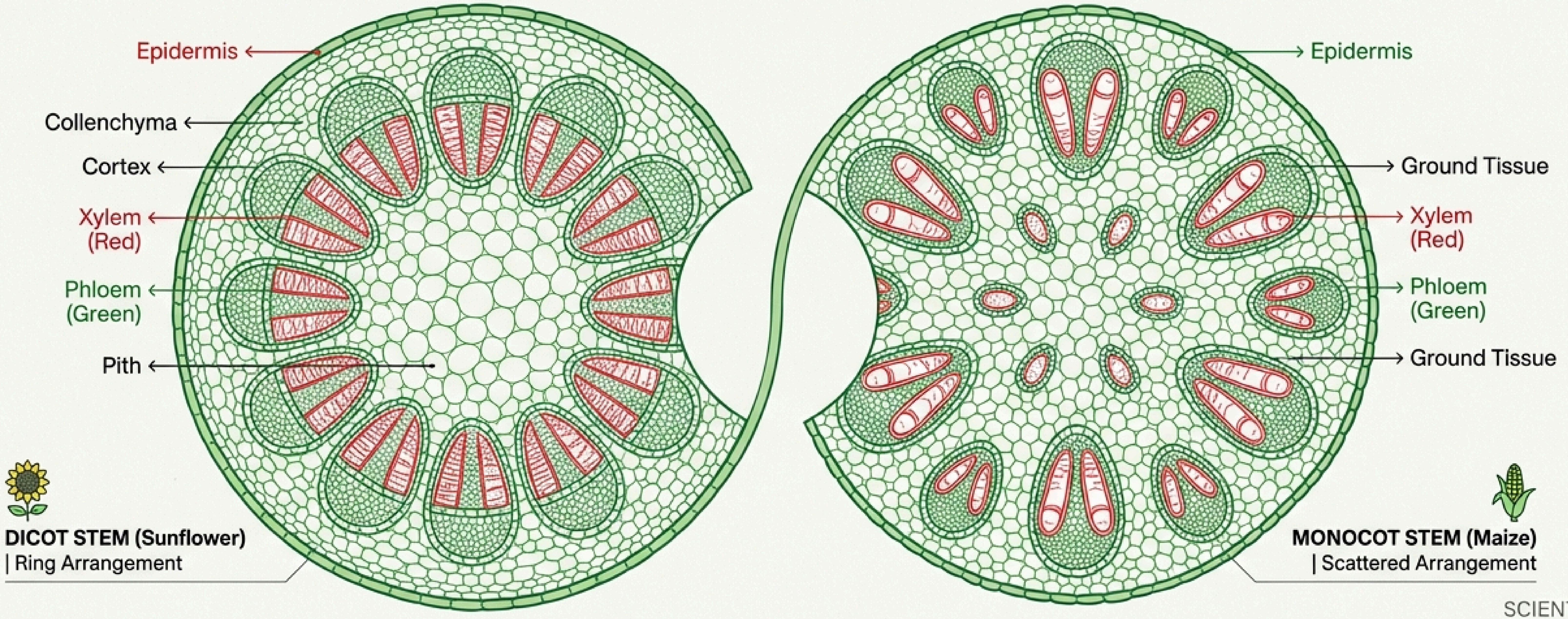
Anatomy of Flowering Plants

Internal Structure & Functional Organization

Class 11 | NEET-UG | Strictly NCERT

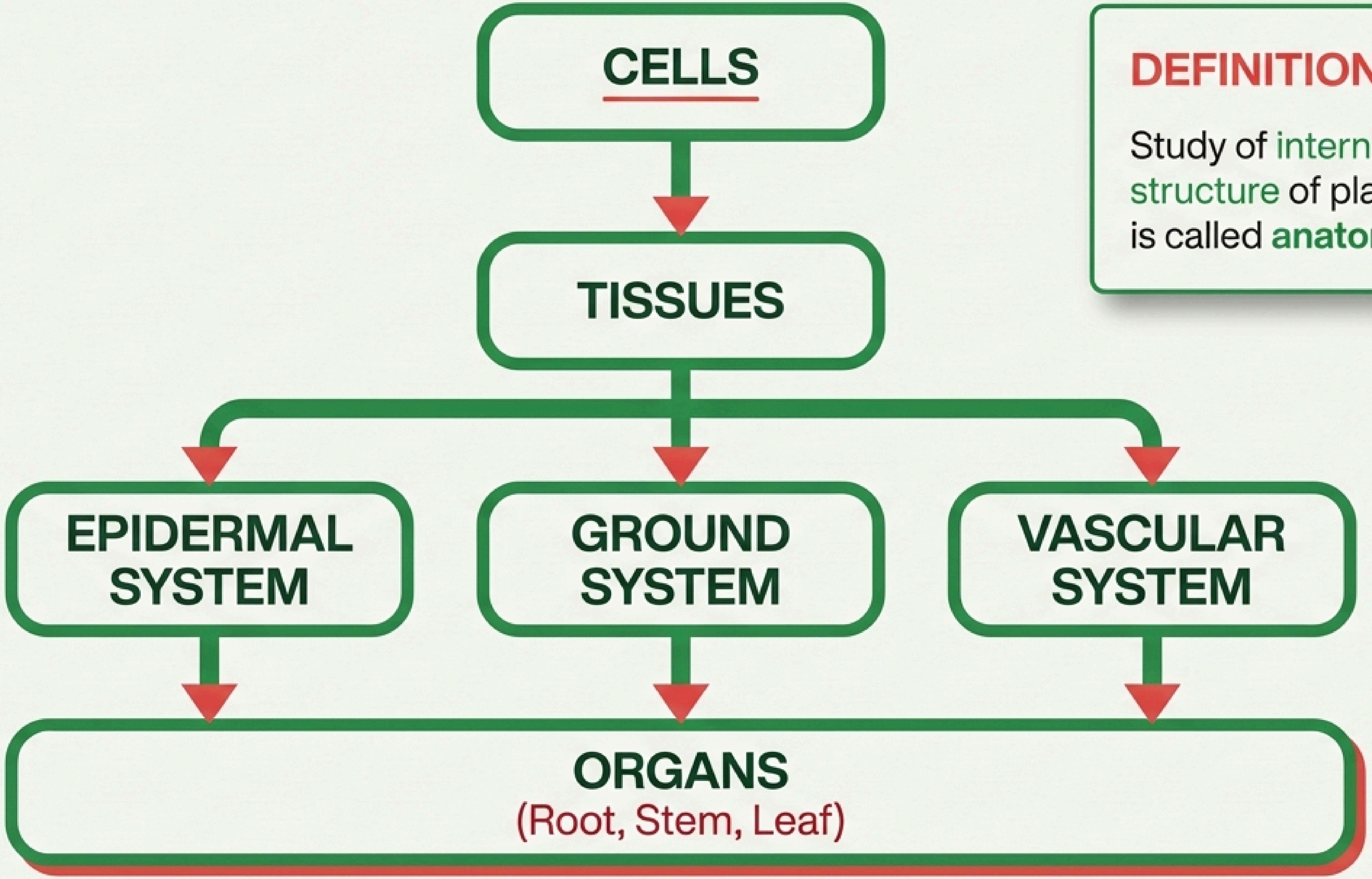
High Weightage: Focus on Diagrams and Differentiation Tables



NCERT

SCIENTIFIC EDITORIAL

SOHAN KUMAR



DEFINITION

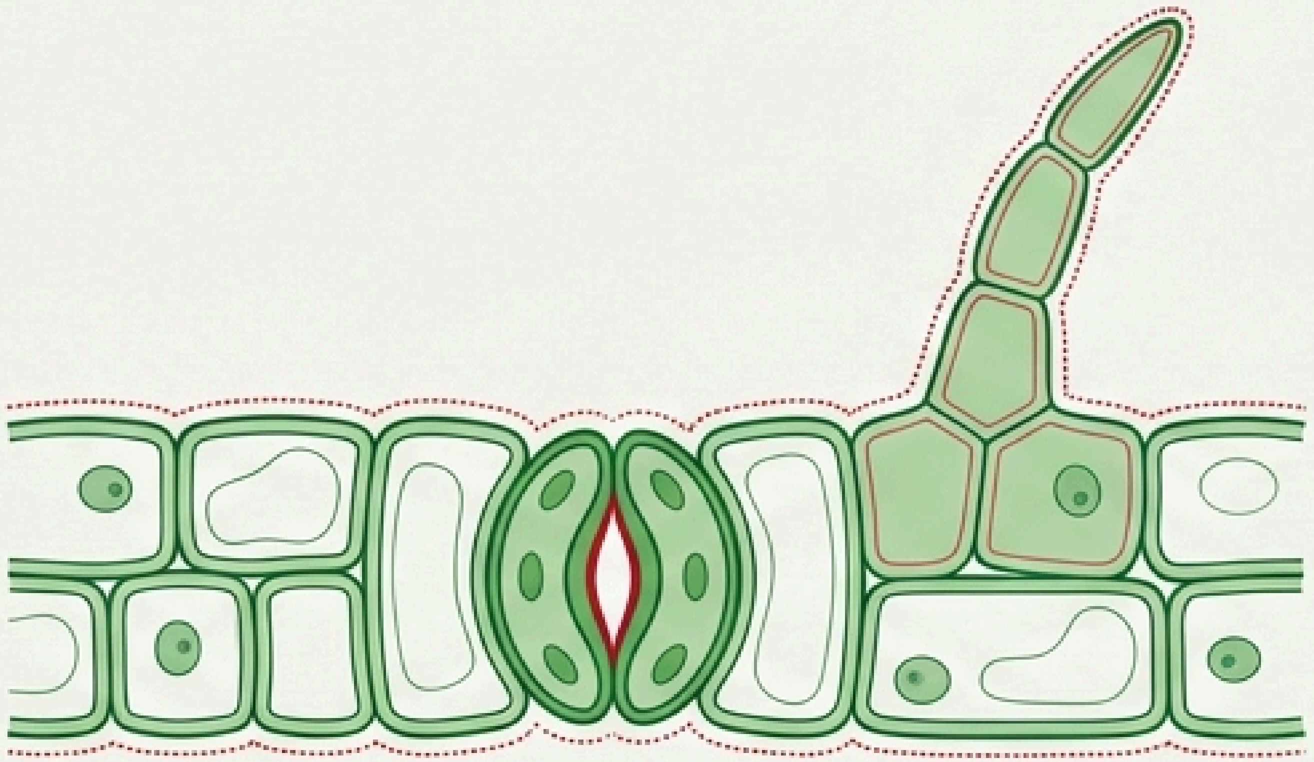
Study of internal structure of plants is called **anatomy.**

SYLLABUS MAPPING

- 6.1 The Tissue System
- 6.2 Anatomy of Dicot & Monocot Plants

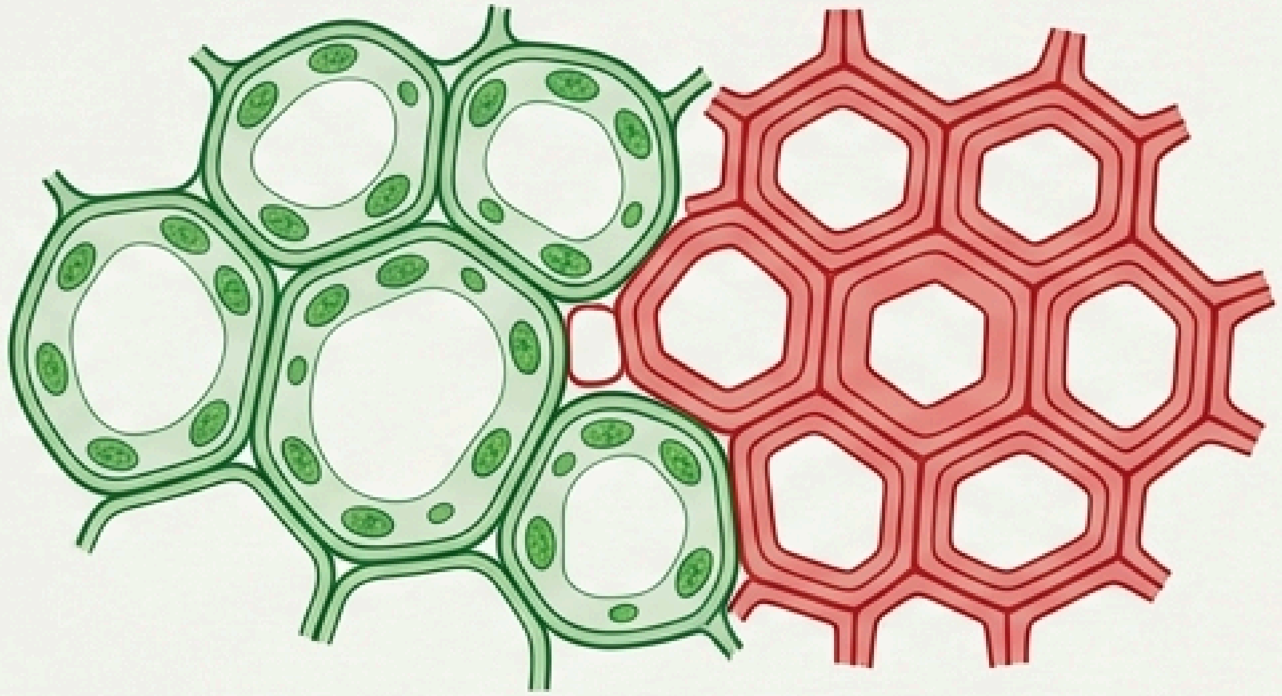
The Tissue Systems

1. Epidermal Tissue System (The Skin)



- Forms the outer-most covering.
- **Components:** Epidermal cells, Stomata, Trichomes/Root Hairs.
- **Cuticle:** Waxy layer preventing water loss (Absent in roots).
- **Trichomes:** Multicellular on stems. Prevent water loss.
- **Root Hairs:** Unicellular. Absorb water & minerals.

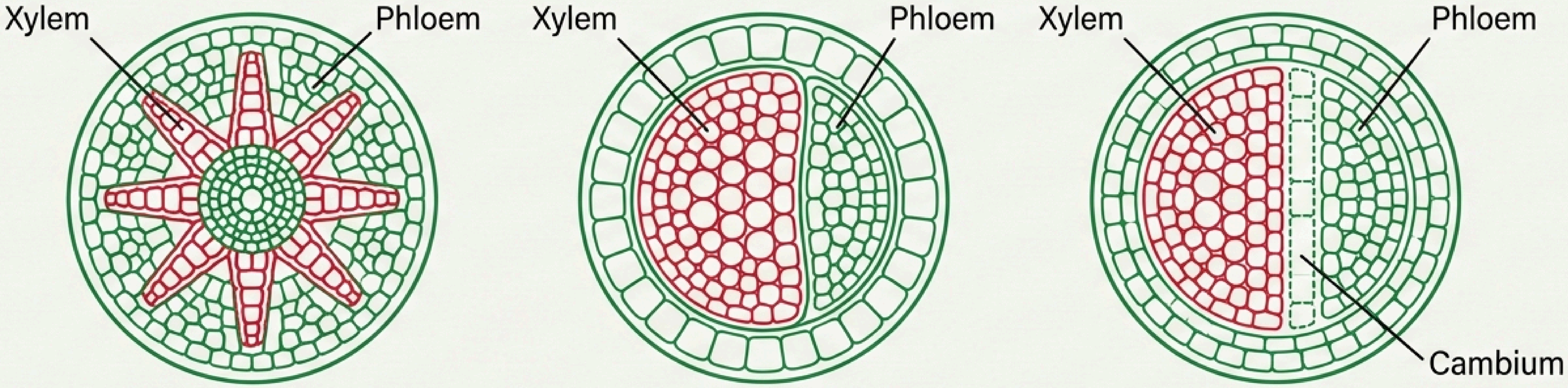
2. Ground Tissue System (The Flesh)



- **Definition:** All tissues except epidermis and vascular bundles.
- **Composition:** Simple tissues (Parenchyma, Collenchyma, Sclerenchyma).
- **Locations:** Cortex, Pericycle, Pith, Medullary rays.
- **In Leaves:** Called **Mesophyll** (Chloroplast-containing).

The Vascular Tissue System

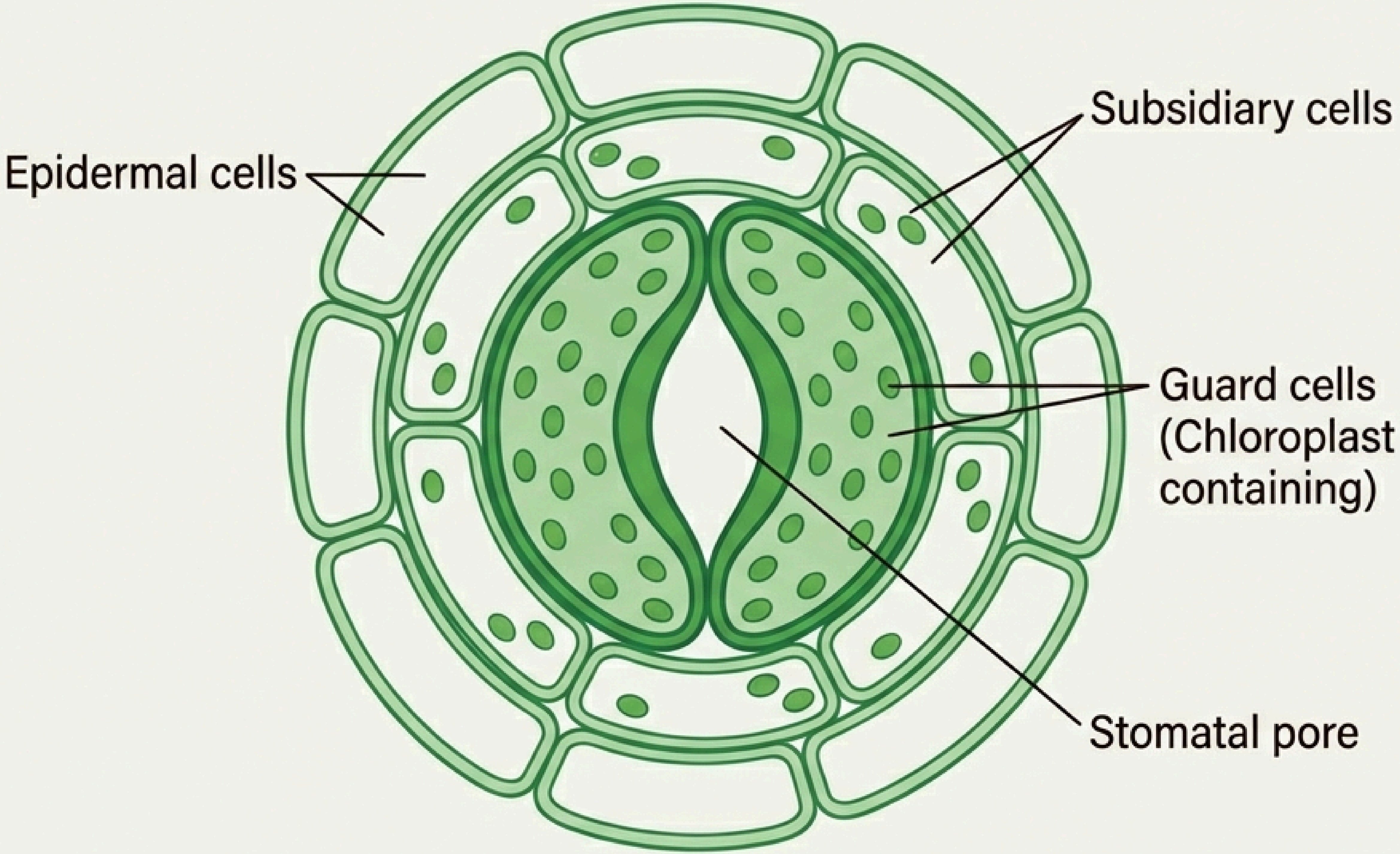
Arrangement of Xylem & Phloem



Radial (Roots) | Conjoint Closed (Monocots) | Conjoint Open (Dicots)

Radial: Xylem & Phloem on different radii.
Conjoint: Xylem & Phloem on same radius (Stems/Leaves).
Open: Cambium present → Secondary Growth.
Closed: No Cambium → No Secondary Growth.

Stomatal Apparatus



Shape Differentiation

Dicots: Bean-shaped guard cells.

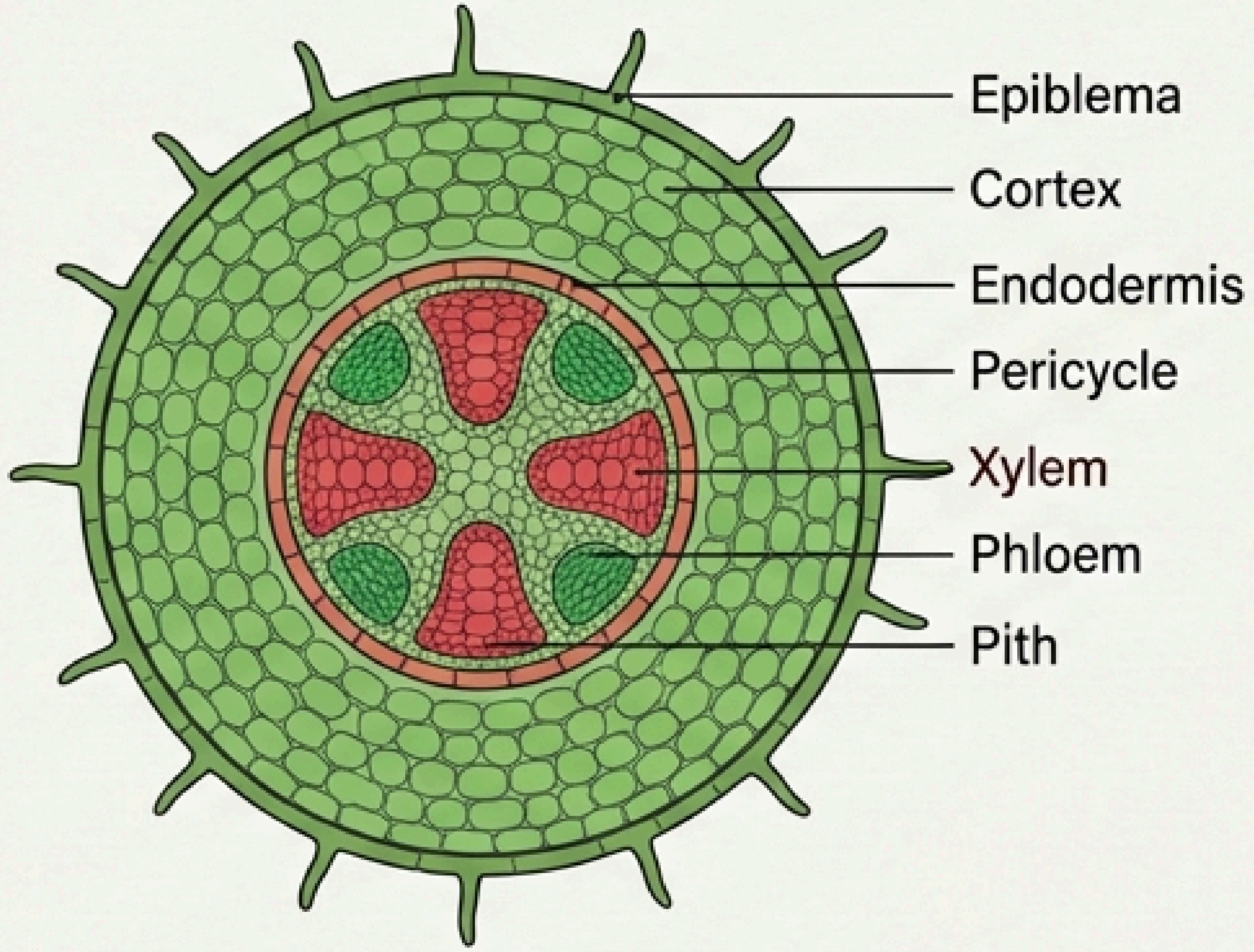
Grasses (Monocots): Dumb-bell shaped guard cells.

Function: Regulate transpiration and gas exchange.

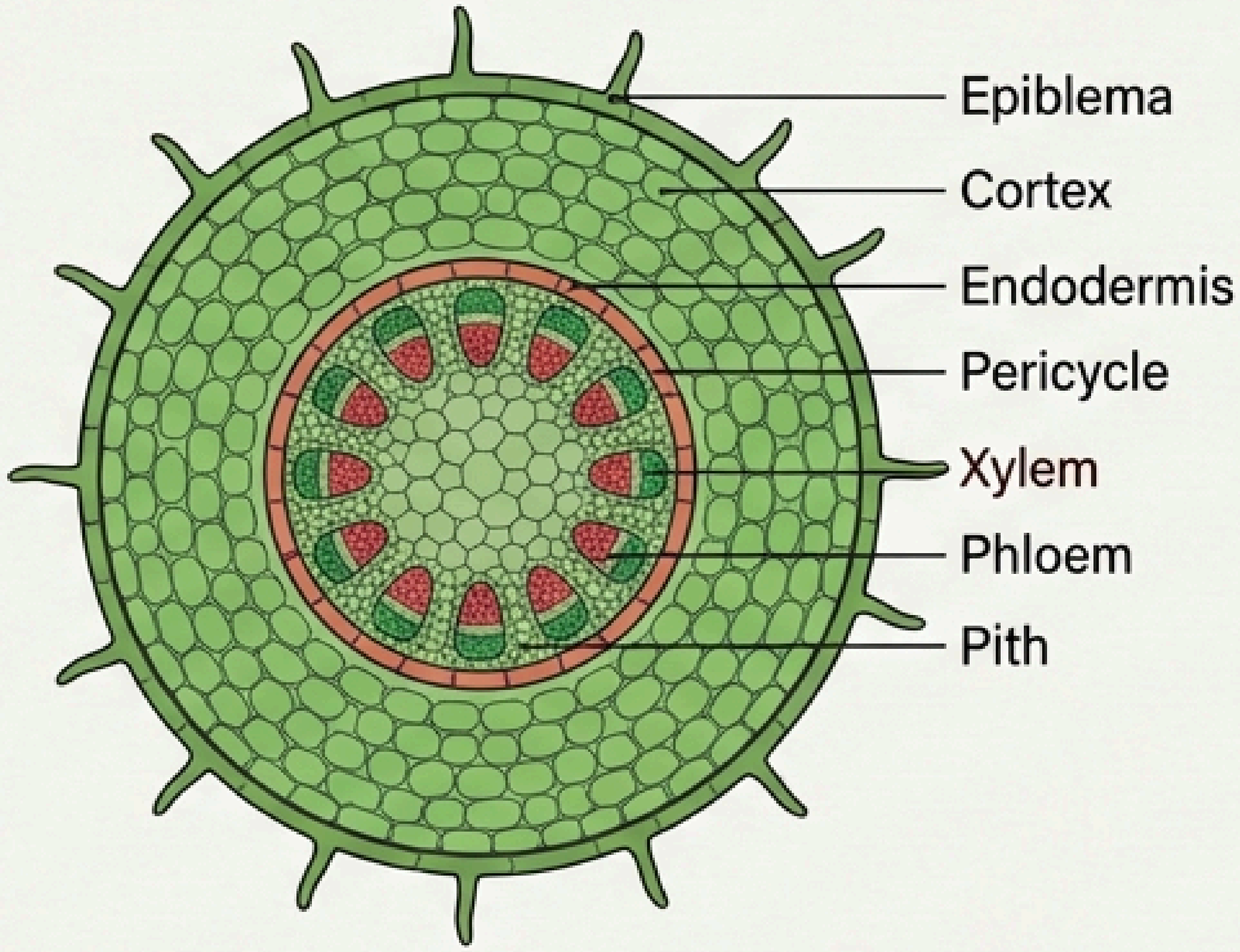
Anatomy of Roots

Comparative Analysis: Dicot vs. Monocot

Dicot Root (Sunflower)



Monocot Root



Dicot Root

Vascular Bundles: Diarch to Hexarch (2-6).
Pith: Small or inconspicuous.
Secondary Growth: Possible.

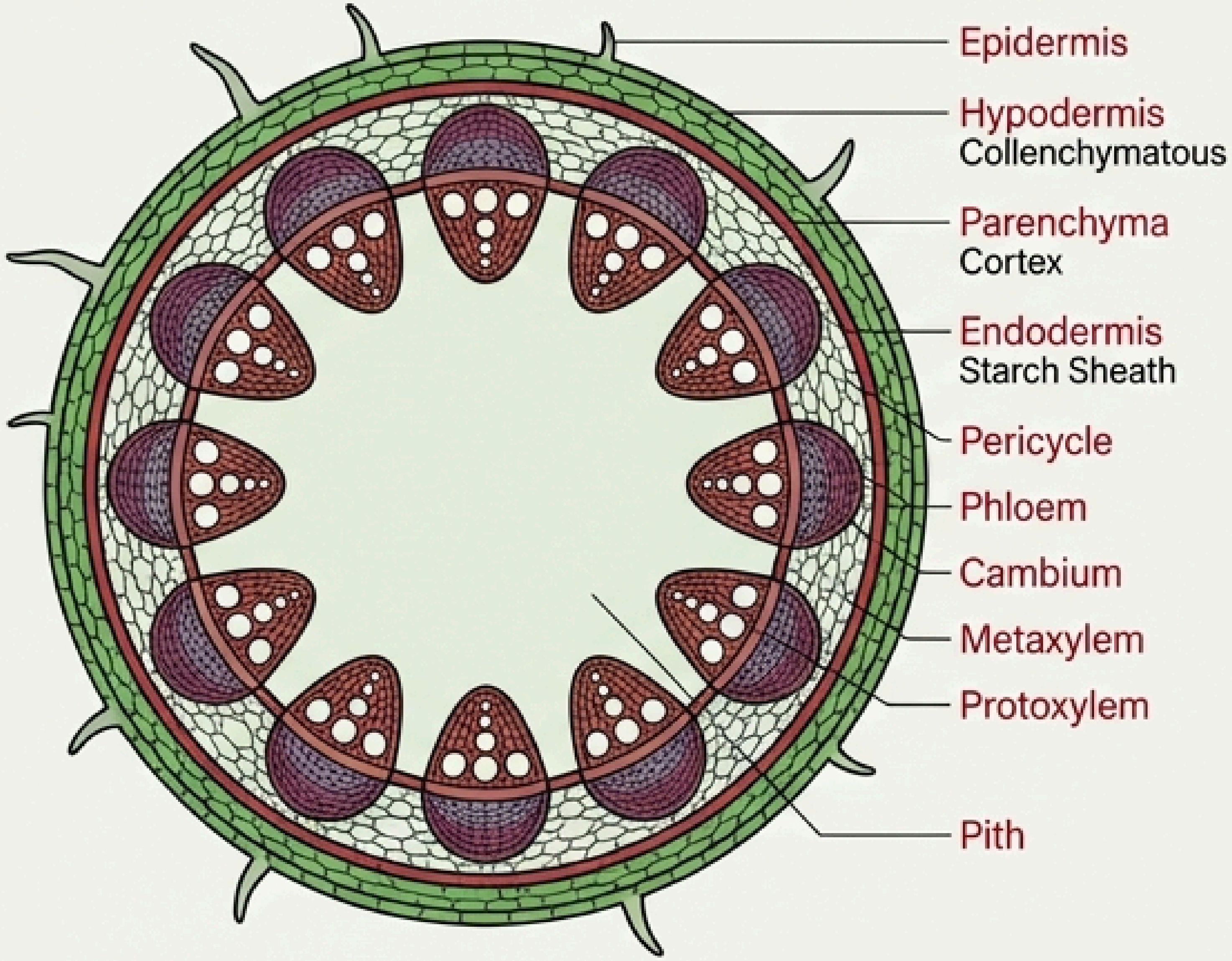
Monocot Root

Vascular Bundles: Polyarch (Usually >6).
Pith: Large and well-developed.
Secondary Growth: Absent.

Common Features: Epiblema (with root hairs), Cortex (Parenchyma), Endodermis (with Casparian Strips), Pericycle.

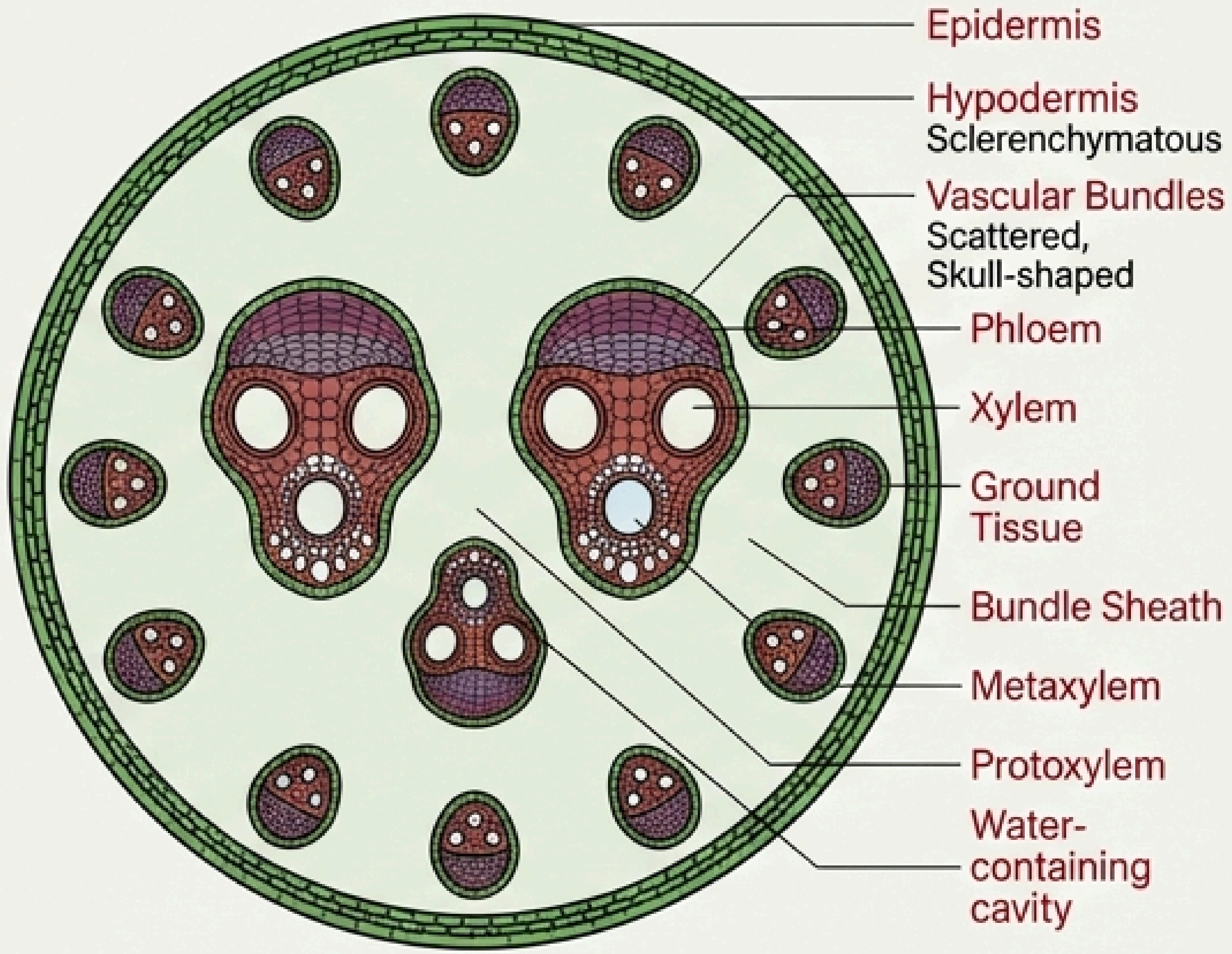
Anatomy of Stems
Critical Exam Topic: Ring vs. Scattered

Dicot Stem



Arrangement: Ring (Characteristic).
Hypodermis: Collenchymatous.
Endodermis: Starch Sheath.
Bundle Type: Conjoint, Open.

Monocot Stem

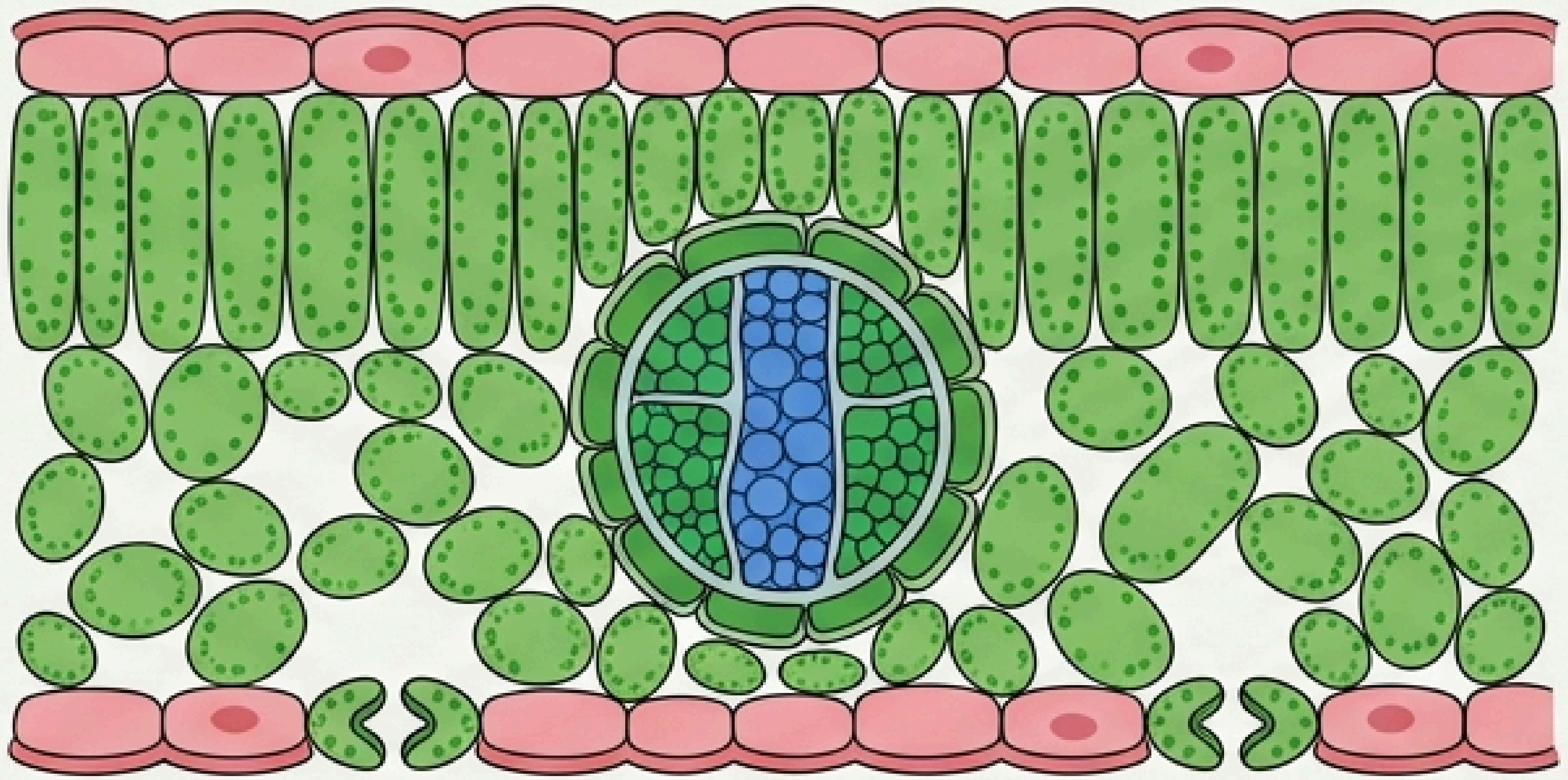


Arrangement: Scattered.
Hypodermis: Sclerenchymatous.
Bundle Sheath: Sclerenchymatous (surrounds bundle).
Unique: Water-containing cavities in V.B.

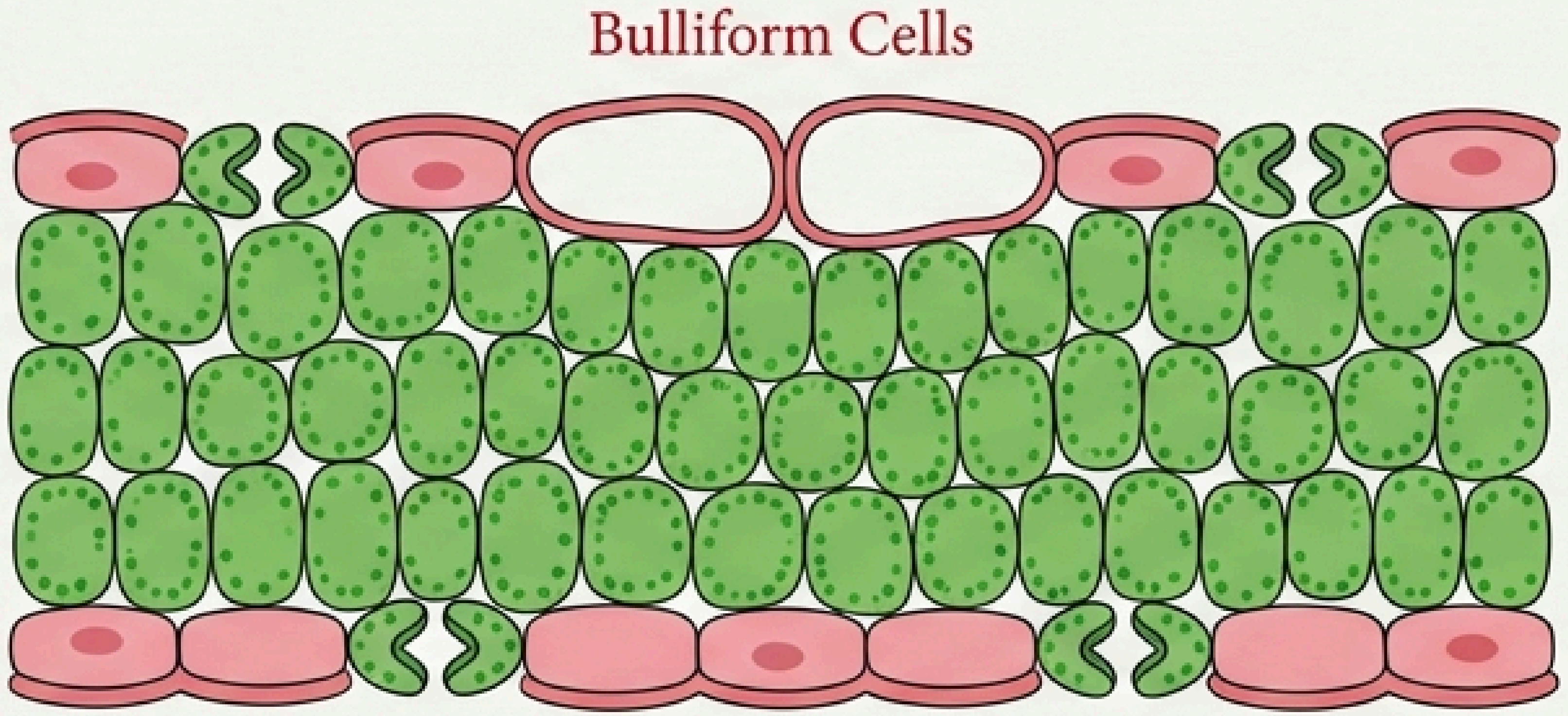
Anatomy of Leaf

Dorsiventral vs. Isobilateral

Dorsiventral Leaf (Dicot)



Isobilateral Leaf (Monocot)



Dicot: Mesophyll differentiated into Palisade & Spongy.

Monocot: Mesophyll undifferentiated. Bulliform cells present (Grasses) for leaf rolling.

Comparative Anatomy: Stem

Feature	Dicot Stem	Monocot Stem
Hypodermis	Collenchymatous	Sclerenchymatous
Ground Tissue	Differentiated (Cortex, Pith)	Undifferentiated Mass
Vascular Bundles	Ring Arrangement	Scattered
Bundle Sheath	Absent	Present (Sclerenchymatous)
Vascular Type	Conjoint, Open	Conjoint, Closed
Phloem Parenchyma	Present	Absent
Water Cavities	Absent	Present in V.B.

⚠ NCERT Line Emphasis

Direct NEET Pickups - Memorize Verbatim

> ‘The tangential as well as radial walls of the endodermal cells have a deposition of water-impermeable, waxy material ‘suberin’ in the form of ‘casparian strips’.’

> ‘In grasses, the guard cells are ‘dumb-bell shaped’.’

> ‘The ‘ring’ arrangement of vascular bundles is a characteristic of ‘dicot stem’.’

> ‘Peripheral vascular bundles are generally ‘smaller’ than the centrally located ones [in Monocot stem].

> ‘Trichomes help in preventing water loss due to transpiration’.

NEET Focus: Concepts & Traps

The Stele Trap

❓ **Question:** What constitutes the stele?

→ **Answer:** All tissues inner to the endodermis.

Stele = **Pericycle** + **Vascular Bundles** + **Pith**

Xylem Orientation

Exarch: Protoxylem towards periphery **(ROOTS)**

Endarch: Protoxylem towards centre **(STEMS)**

Location Confusion

- **Casparian Strips** → Endodermis of Roots
- **Starch Sheath** → Endodermis of Dicot Stem
- **Bulliform Cells** → Adaxial Epidermis of Monocot Leaves

Assertion (A) & Reason (R) Analysis

Card 1

Assertion: "Vascular bundles in monocot stems are considered closed."

Reason: "They lack cambium and cannot form secondary tissues."

Result: Both True. Reason is the Correct Explanation.

Card 2

Assertion: "The trichomes in the shoot system help in preventing water loss."

Reason: "They are usually multicellular and may be secretory."

Result: Both True. Reason is NOT the correct explanation.

Card 3

Assertion: "Pith is well developed in Dicot roots."

Reason: "Monocot roots have polyarch vascular bundles."

Result: Assertion is False. Reason is True.

Rapid Revision Keywords

Polyarch

>6 Xylem bundles (Monocot Root)

Diarch to Hexarch

2–6 patches (Dicot Root)

Conjoint

Xylem/Phloem on same radius (Stem/Leaf)

Bulliform Cells

Motor cells for leaf rolling (Grasses)

Pericycle Function

Lateral roots & Cambium formation (Dicot Root)

Mesophyll

Photosynthetic ground tissue of leaf

Final Chapter Snapshot

The Anatomy Matrix

	General Feature	Dicot Specifics	Monocot Specifics
ROOT	• Radial Bundles, Exarch Xylem.	• Diarch-Hexarch, Small Pith, Secondary Growth.	• Polyarch, Large Pith, No Secondary Growth.
STEM	• Conjoint Bundles, Endarch Xylem.	• Ring Arrangement, Open, Collenchyma Hypodermis.	• Scattered, Closed, Sclerenchyma Hypodermis, Water Cavities.
LEAF	• Mesophyll Ground Tissue.	• Dorsiventral, Differentiated Mesophyll (Palisade/Spongy).	• Isobilateral, Undifferentiated Mesophyll, Bulliform Cells.